

Amr Waled

Mechanical Power Engineering Student — 3rd Year | Tanta University

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Tanta, Egypt

EDUCATION

B.Sc. in Mechanical Power Engineering

2022 — Present

Faculty of Engineering, Tanta University — New Regulation

Currently in 3rd Year, 2nd Semester

- Relevant Coursework: Thermodynamics, Fluid Mechanics, Heat Transfer, Gas Dynamics, Internal Combustion Engines, Fuel Systems, Air Conditioning & Refrigeration, Power Systems, Renewable Energy Systems, Engineering Drawing

TECHNICAL SKILLS

CAD & Simulation: SolidWorks (Intermediate), AutoCAD (Intermediate), ANSYS Fluent / Workbench (Intermediate)

Engineering: CFD Analysis, HVAC Load Calculations, Duct Sizing, Fuel System Analysis, 3D Parametric Modeling

Programming & Tools: Python (Basic), Arduino (Intermediate), Microsoft Office (Advanced), Google Sheets (Advanced)

Other: Media Buying (Proficient), AI Prompting & Vibe Coding (Advanced), Technical Presentations

CFD Simulation of Supersonic Flow over a Double-Wedged Airfoil

Gas Dynamics Course

Tools: ANSYS Fluent, ANSYS Workbench

- Performed numerical simulation of supersonic compressible flow over a double-wedged airfoil at multiple angles of attack using ANSYS Fluent with a density-based solver
- Captured oblique shockwaves, expansion fans, and boundary-layer interactions; analyzed Mach number contours, static pressure, and temperature distributions
- Compared aerodynamic performance across varying attack angles and documented results in a structured technical report

HVAC System Design — Doha Hotel, Qatar (Team Project)

Air Conditioning Course

Scope: Full HVAC design for a multi-floor hotel facility

- Performed detailed cooling and heating load calculations accounting for Qatar's extreme climate conditions
- Designed and sized the complete duct network for the building
- Conducted a feasibility study comparing multiple HVAC system configurations and selected the optimal system based on energy efficiency, capital cost, and operational performance

Fuel Systems Analysis — Perkins New 1000 Series

Fuel Systems Course

- Selected and technically justified the Perkins New 1000 Series diesel engine fuel injection system following a comparative analysis of available options
- Researched system architecture, injection timing, fuel delivery mechanisms, and performance characteristics
- Compiled and delivered a structured technical presentation to the faculty supervisor

ICE II — Interactive Web Simulator and Problem Solver

Internal Combustion Engines II

Tools: HTML, CSS, JavaScript, AI-Assisted Development

- Built a web application that simulates the fuel-air intake process of an internal combustion engine with interactive user inputs
- Implemented a problem solver covering all major course calculations with custom parameter inputs

Renewable Energy Systems — Web Calculator and Angle Simulator

Renewable Energy Course

Tools: HTML, CSS, JavaScript, AI-Assisted Development

- Developed an interactive web tool for all renewable energy coursework calculations and a visual simulation of solar panel orientation angles vs. energy output

4-Cylinder Engine Model — SolidWorks

Mechanical Design

- Designed a full 4-cylinder engine assembly including pistons, crankshaft, camshaft, and cylinder block using parametric 3D modeling and assembly constraints in SolidWorks

Obstacle Avoidance Autonomous Robot — Arduino

Electronics Project

- Built an autonomous robot using Arduino and ultrasonic sensors for real-time obstacle detection; programmed path decision logic and motor control in embedded C

TECHNICAL TRAINING

Industrial Training — Talakha Thermal Power Station

Summer 2025 | 3 Weeks

Egyptian Electricity Holding Company, Dakahlia, Egypt

- Gained hands-on exposure to steam turbines, boilers, generators, and power distribution systems within a live thermal power plant environment
- Observed and documented standard operational and maintenance procedures

SolidWorks Intensive Training — NOVA CAMP

Summer 2025 | 1 Month

Mansoura University

- Completed intensive training in 3D mechanical design, parametric modeling, assemblies, and engineering drawings using SolidWorks

WORK EXPERIENCE

Marketing Manager

April 2026 — Present

Engaz Developments — Real Estate

- Managing full-funnel digital marketing campaigns and paid media buying strategies
- Developing data-driven marketing plans to drive qualified lead generation for real estate projects

Business Development Intern

2025

AIBootcamp — EdTech

- Supported business growth and partnership development within an EdTech startup
- Conducted market research and assisted in identifying new client acquisition opportunities

LEADERSHIP & ACTIVITIES

Class Representative — Mechanical Power Engineering

2022 — 2024 | 2 Years

Faculty of Engineering, Tanta University

- Elected class representative for the Mechanical Power Engineering cohort for two consecutive years
 - Served as the primary point of communication between students and faculty, coordinating academic matters, exam schedules, and departmental issues
 - Facilitated conflict resolution and maintained a productive learning environment for the batch
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President — Enactus Tanta University Chapter

2025 — Present

Enactus Egypt

- Leading chapter operations and social entrepreneurship projects at the national level
 - Founded and organized Xplore Camp — a career development program covering CV building, LinkedIn, Marketing, Presentations, and Entrepreneurship
 - Secured a strategic partnership with V7 to co-execute the camp, enhancing program reach and quality
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Public Relations — IEEE Tanta Student Branch

2022 — 2024

- Represented the branch in external communications, events promotion, and industry outreach
 - Helped organize engineering workshops and technical events for faculty students
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Public Relations — Rally & Insider Student Activities

2022 — 2024

- Managed communications and partnerships for two active student organizations, driving event attendance and visibility
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LANGUAGES

Arabic: Native

English: Professional Working Proficiency